

SIMATS ENGINEERING

ASSESSMENT TOOL 1 : Concept Mapping

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I SRINIVAS S/192524129 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SRINIVAS

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect

		accurate and meaningful			
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Permitted Tools :

draw.io / diagrams.net, Lucidchart, MindMeister, Miro, Canva, PowerPoint, Google Drawings, or equivalent diagramming tool

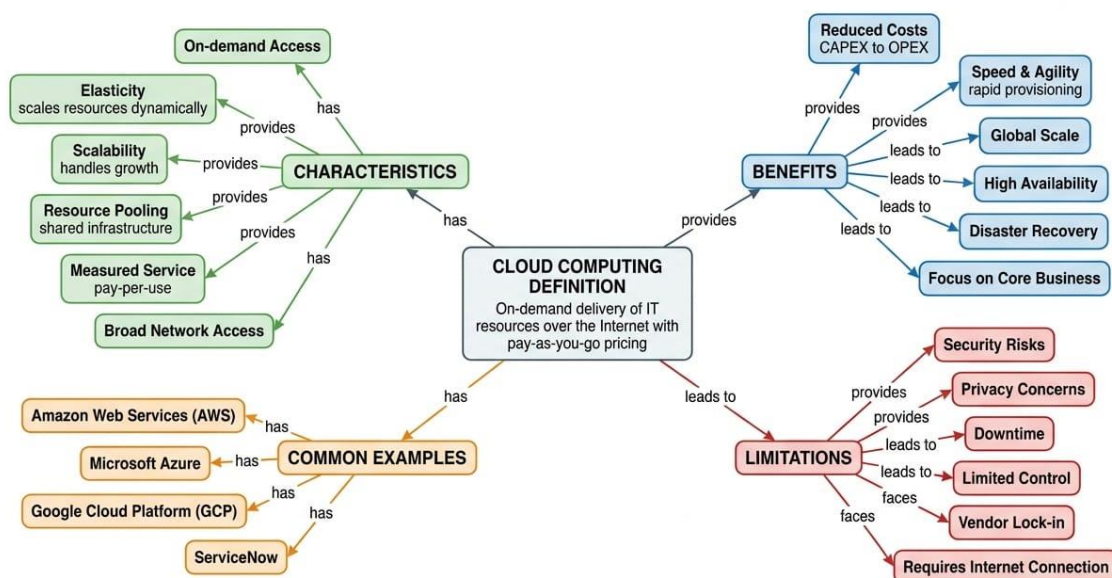
Procedure for Preparing the Digital Concept Map

1. Use the following procedure to prepare the diagram and upload evidence.
2. Students who already know another diagramming tool may use it, but the exported evidence and source file must still be submitted.
3. Open <https://app.diagrams.net/> or the desktop version of draw.io / diagrams.net.
4. Choose Device, Google Drive, or OneDrive as the save location according to availability.
5. Create a blank diagram and name the file as
CSA15_CO1_AT1_RegisterNumber_Concept_Map.drawio.
6. Place the concept map title at the center or as a clearly visible anchor node.
7. Add major cloud nodes: cloud definition, cloud evolution, cloud architecture, service models, deployment models, virtualization, web services, SOAP, REST, APIs, elasticity, scalability, resource pooling, measured service, and emerging paradigms.
8. Connect nodes with labeled arrows such as uses, depends on, enables, scales through, communicates through, belongs to, supports, provides, consumes, or controls.
9. Use colors meaningfully: one color for service models, one for cloud characteristics, one for API/web-service concepts, and one for concept map title-specific elements.
10. Export the final diagram as PNG or PDF. Keep the editable .drawio source file also.
11. Paste the exported diagram image or screenshot in this DOCX under Task 1.
12. Upload the DOCX, exported PNG/PDF, and .drawio file to Google Classroom.

Task 1: Concept Map

Create a concept map in draw.io, Lucidchart, Miro, Figma, or any suitable diagramming tool. The map must connect cloud ecosystem, cloud actors, service models, deployment models, virtualization, web services, REST/SOAP APIs, scalability, elasticity, and the student's Project.

Concept Map 1: Cloud Computing Fundamentals



Task 2A: Service Model Responsibility Matrix

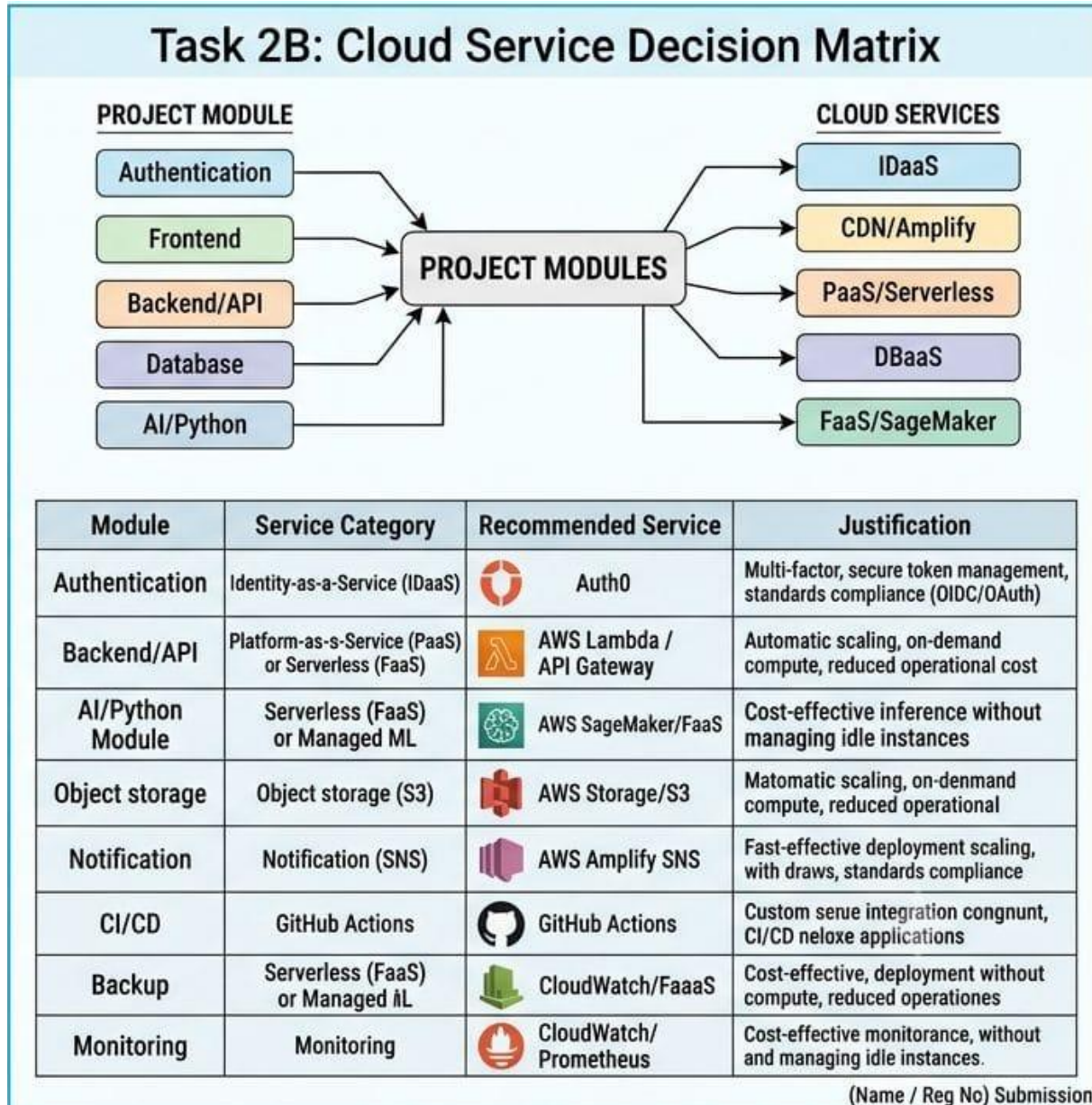
Prepare a technical matrix that compares SaaS, PaaS, IaaS, FaaS/serverless, database-as-a-service, storage-as-a-service, identity-as-a-service, and monitoring/logging services. For each selected service category, explain what is managed by the cloud provider, what is managed by the student application team, and why that split fits the Project.

Task 2A: Service Model Responsibility Matrix

Service Category	Cloud Provider Managed	Student Team Managed	Project Justification
 IaaS (Azure VMs)	• datacenter, • physical servers, • networking, storage	• OS, • middleware, • application	• Custom kernel required for specific performance needs, full control needed
 PaaS (AWS Elastic Beanstalk)	• physical inf, OS, • runtime	• application, data	• Fast application deployment without managing OS patching
 SaaS (Google Workspace)	• everything from infra to application	• user management, • settings	• Standard tools like email and document editing with minimal config
 FaaS/Serverless (AWS Lambda)	• server, runtime, • networking, scalability	• function code, triggers	• Ideal for handling varying, decoupled workloads without constant compute
 Database-as-a-Service	• Database-as- • auto-as-a-storage	• application, application, data	• Automatic tools like email, and document toods with mintrnal config
 Storage-as-a-Service	• Application • networking • storage storage	• data, applications	• Fast, on-managing and comput without managing project patching
 Identity-as-a-Service	• User management, and nanction	• user managemt, managent, • settings	• Standard tools like email and document editing with minimal config
 Monitoring/Logging Services	• Datacenter, datacenter • physical servers, • storage • Storage-as--networking, Loggina	• finction code • Scability management • User management	• Ideal for allication deployment without managing OS patching • Fast applicatios like email and memicisem editing and minimal config:

Task 2B: Cloud Service Decision Matrix

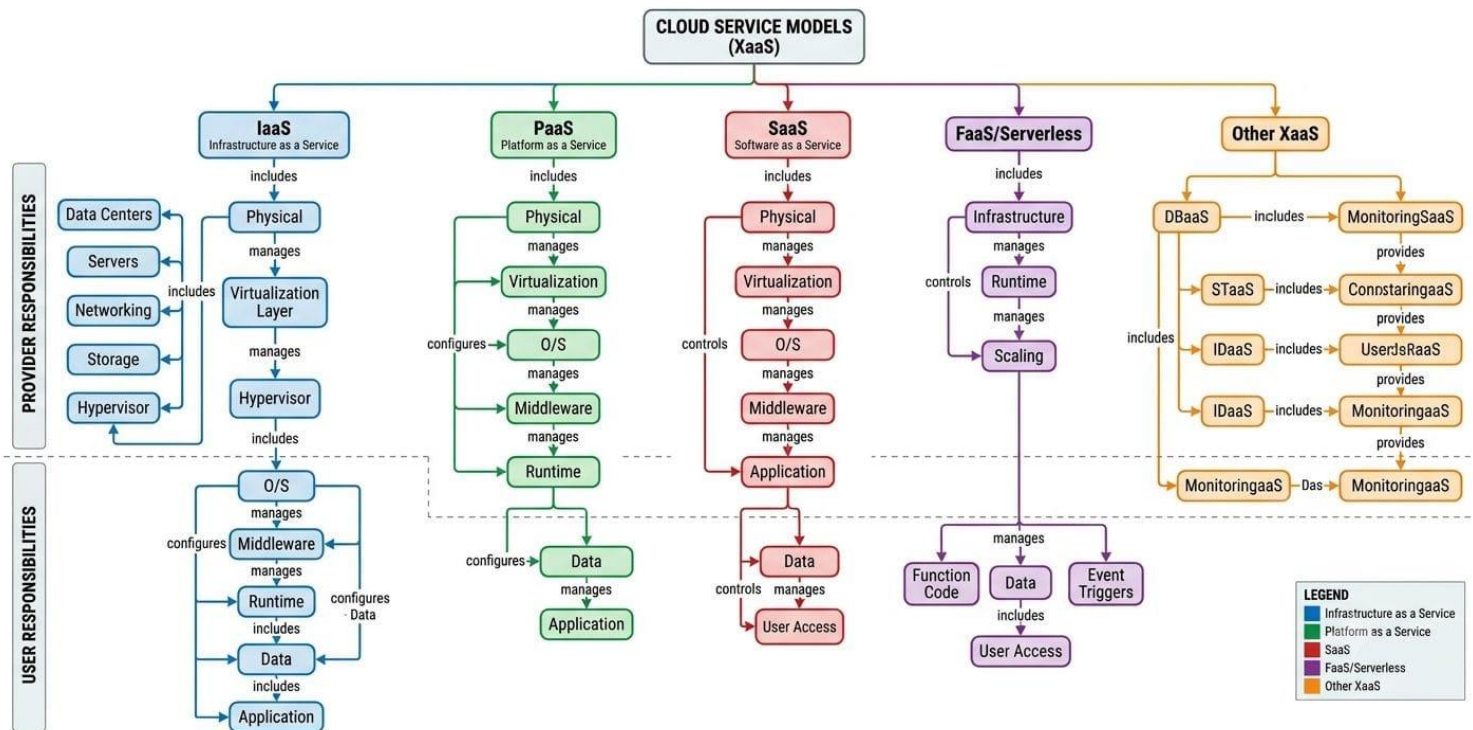
Map Project modules to cloud services. Include authentication, frontend hosting, backend/API layer, database, object storage, analytics/logging, notification, AI/Python module, security, CI/CD, backup, and monitoring. Students should justify the most relevant service choices instead of forcing every possible cloud service into the design.



Task 3: API Flow, Elasticity, and Scalability Design

Draw a professional cloud-flow diagram for the Project modules. The diagram must show user app or web app, API gateway/backend service, authentication, compute layer, database, object storage,

analytics/logging, monitoring, security boundaries, and scaling points. Students may replicate the given visual pattern, but all components and labels must be customized to their own architecture.



Task 4: Technical Justification

Write a concise justification explaining how the diagram and matrices prove CO1. The justification must discuss service model selection, API role, elasticity, scalability, virtualization or managed compute choice, and cloud responsibility boundaries.

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Concept Map 4: Virtualization & Web Services as Cloud Enablers

